PARKTOWN BOYS' HIGH SCHOOL

ARISE

Department of Mathematics

Assignment

21 April 2026

Subject	Mathematics
Grade	12
Topic	Financial Mathematics and Analytical Geometry
Marks	50
Time	1 Hour

Examiner	Mrs S Rowell
Moderator	Ms H Gouws

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This assignment consists of 3 questions, 5 pages and a formula sheet. Please ensure that your paper is complete.
   You may detach your formula sheet for reference.

2. Answer ALL the questions in the answer book provided.

3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.

4. Answers only will NOT necessarily be awarded full marks.

5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.

6. If necessary, round off answers to TWO decimal places, unless stated otherwise.

7. Diagrams are NOT necessarily drawn to scale.

8. Use the mark allocation to guide the length of your solution.

9. Write neatly and legibly.

10. Rule off after each question.

QUESTION 1

1.1 Kyle decides to invest R7 000 at the end of every month to save for a deposit on a house. Interest on the investment account is calculated at 15% p.a. compounded monthly.

1.1.1 Calculate the effective annual rate on the investment. (2)

1.1.2 What will the value of the investment be at the end of four years? (2)

1.1.3 Determine the minimum number of months required for the investment to reach a value greater than R1 000 000. (4)

1.2 Joyce pays off a loan of R45 000 by means of equal quarterly payments, starting 1 year after the loan is granted and ending 4 years after the loan is granted. The interest rate is 9% p.a. compounded quarterly.

1.2.1 Calculate Joyce's quarterly payment. (4)

1.2.2 Calculate the total interest that Joyce will pay on the loan. (2)

1.2.3 Determine the outstanding balance on the loan directly after the 6^{th} payment. (4)

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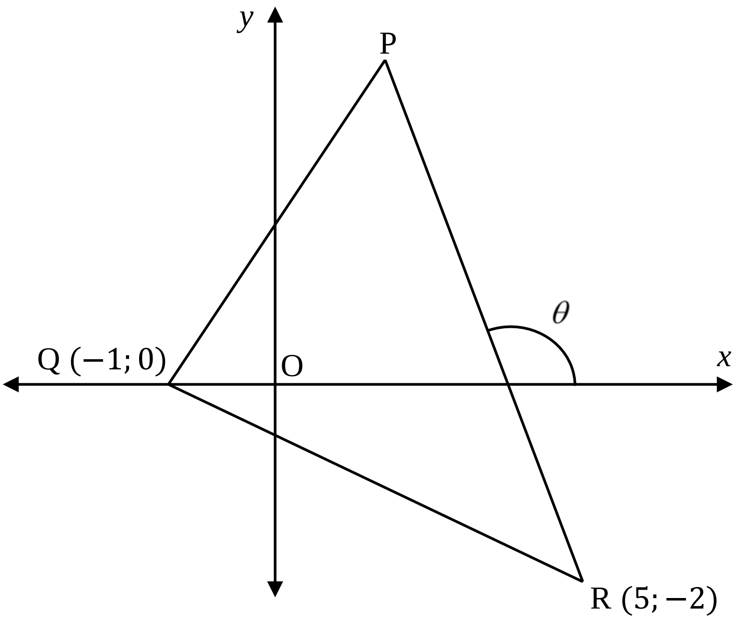
QUESTION 2

In the diagram below, P, Q $(-1; 0)$ and R $(5; -2)$ are vertices of triangle PQR such that the angle of inclination of PR is θ. The equation of PQ is given as $3x - y + 3 = 0$ and the equation of PR is given as $y = -2x + 8$.



2.1 Calculate the length of QR. Leave your answer in simplest surd form. (2)

2.2 Prove that $\mathrm{P\hat{Q}R} = 90°$. (3)

2.3 Calculate the value of θ. (2)

2.4 Calculate the coordinates of P. (3)

2.5 Determine the equation of the circle passing through P, Q and R in the form $(x-a)^2 + (y-b)^2 = r^2$. (4)

2.6 The centre of another circle also lies on the line PR. If this circle passes through Q and $(8;3)$, determine the coordinates of the centre of the circle. (5)

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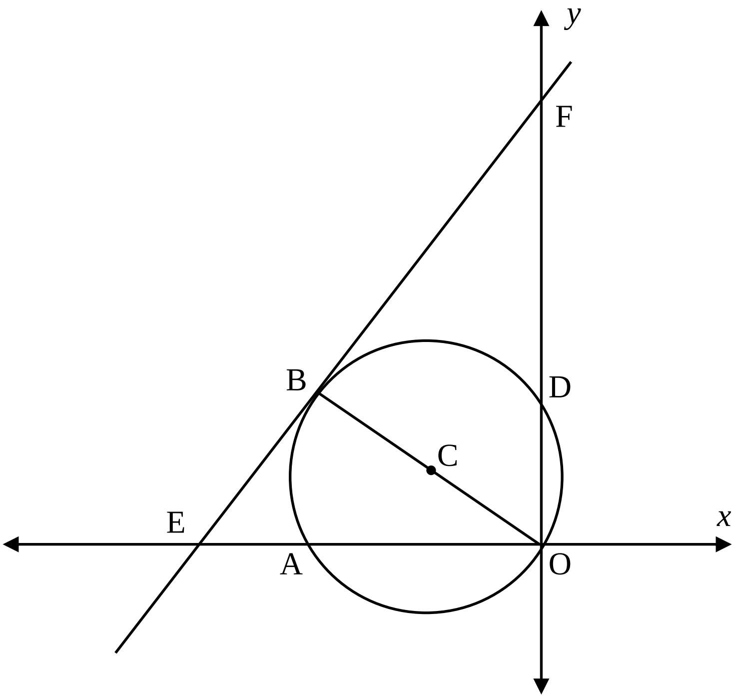
QUESTION 3

A circle with centre at C has the equation $x^2 + y^2 + 8x - 6y = 0$. The circle passes through the origin and intersects the x and y- axes at A and D respectively.
EF is a tangent to the circle at B.
EF intersects the x and y- axes at E and F respectively.



3.1 Show that the coordinates of C are $(-4; 3)$. (2)

3.2 Determine the coordinates of B. (2)

3.3 Determine the equation of the tangent EF in the form $y = mx + c$. (4)

3.4 It is further given that the length of BF is $\frac{40}{3}$ units.
Calculate the area of ΔOBF. (3)

3.5 Determine the value(s) of k such that the y-axis is a tangent to the circle $(x-k)^2 + (y-3)^2 = 25$. (2)

[13]